Practical – 9

Title :- **Creating a Cube in SQL server and Developing an OLAP Cube**

Creating Data Warehouse and Develope an OLAP Cube.

Theory :-

A] **Data Staging Area**

- **Definition**: An intermediate storage zone used during the **ETL (Extract, Transform, Load)** process.
- **Purpose**:
  - Collects raw data from multiple sources.
  - Cleans, transforms, and integrates data before loading into the data warehouse.
- **Key Role**: Ensures consistency, quality, and readiness of data for analysis. Think of it as a "pre-processing kitchen" before serving data to the warehouse.

B] OLAP Models

Model	Storage	Strengths	Weaknesses	Best Use Case
ROLAP (Relational OLAP)	Data stored in **relational databases** (tables).	Handles very large datasets; leverages SQL; scalable.	Slower query performance; limited by SQL functions.	When data volume is huge and relational DBMS is already in place.
MOLAP (Multidimensional OLAP)	Data stored in **multidimensional cubes**.	Very fast query performance; pre-computed aggregates; intuitive cube navigation.	Consumes more storage; less scalable for massive datasets.	When speed of analysis is critical and dataset size is manageable.
HOLAP (Hybrid OLAP)	Combines **ROLAP + MOLAP**.	Balances scalability and speed; stores detailed data in relational DB, aggregates in cubes.	More complex architecture; can be harder to maintain.	When both scalability and fast query response are needed.

C] Quick Analogy

- **ROLAP**: Like querying a giant library catalog (flexible, but slower).
- **MOLAP**: Like flipping through a pre-organized index (super fast, but limited by cube size).
- **HOLAP**: Like having both the catalog and the index — best of both worlds, but requires careful setup.

Procedure

[1] Creating Data Warehouse

1. Open SQL Server Management Studio 2008
2. Connect Database Engine
3. Open **New Query** editor
4. Copy paste Scripts given below in various steps in new query editor window one by one
5. To run the given SQL Script, press F5
6. It will create and populate "Sales_DW" database on your SQL Server

[2] **Developing an OLAP Cube**

For creation of OLAP Cube in Microsoft BIDS Environment, follow the 10 easy steps given below.

Step 1: Start BIDS Environment

Click on **Start Menu** -> Microsoft **SQL Server 2008 R2** -> Click **SQL Server Business Intelligence Development Studio.**

Step 2: Start Analysis Services Project

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Click **File** -> **New** -> **Project** ->**Business Intelligence Projects** ->select **Analysis Services Project** -> Assign Project **Name** (SalesDataAnalysis) -> Click **OK**

Step 3: Creating New Data Source

3.1 In Solution Explorer, Right click on **Data Source** -> Click **New Data Source**

3.2 Click on Next

3.3 Click on New **Button**

3.4 Creating **New connection**

a) Specify Your SQL **Server Name** where your Data Warehouse was created
b) Select Radio Button according to your **SQL Server Authentication** mode
c) Specify your **Credentials** using which you can connect to your SQL Server
d) Select database Sales_DW.
e) Click on **Test Connection** and verify for its success
f) Click OK.

3.5 Select Connection created in **Data Connections**-> Click **Next**

3.6 Select Option **Inherit**

3.7 Assign Data Source **Name** -> Click **Finish**

Step 4: Creating New Data Source View

4.1 In the Solution Explorer, Right Click on **Data Source View** -> Click on **New Data Source View**

4.2 Click **Next**

4.3 Select **Relational Data Source** we have created previously (Sales_DW)-> Click **Next**

4.4 First move your **Fact Table** to the right side to include in object list.
Select FactProductSales Table -> Click on Arrow Button to move the selected object to Right Pane.

4.5 Now to **add dimensions** which are **related** to your **Fact Table**, follow the given steps:
Select **Fact Table** in Right Pane (Fact product Sales) -> Click On **Add Related Tables**

4.6 It will add all associated dimensions to your Fact table as per relationship specified in your SQL DW (Sales_DW). Click Next.

4.7 Assign **Name (SalesDW DSV)->** Click **Finish**

4.8 Now Data Source View is ready to use.

Step 5: Creating New Cube

5.1 In Solution Explorer -> Right Click on **Cube->** Click **New Cube**

5.2 Click **Next**

5.3 Select Option **Use existing Tables ->** Click **Next**

5.4 Select Fact Table Name from **Measure Group Tables (FactProductSales) ->** Click **Next**

5.5 Choose **Measures** from the List which you want to place in your Cube --> Click **Next**

5.6 Select All **Dimensions** here which are associated with your Fact Table-> Click **Next**

5.7 Assign **Cube Name** (SalesAnalyticalCube) -> Click **Finish**

5.8 Now your Cube is ready, you can see the newly created cube and dimensions added in your solution explorer.

Step 6: Dimension Modification

In Solution Explorer, double click on dimension **Dim Product ->** Drag and Drop Product Name from Table in Data Source View and Add in Attribute Pane at left side.

Step 7: Creating Attribute Hierarchy In Date Dimension

Double click On **Dim Date** dimension -> Drag and Drop Fields from Table shown in Data Source View to Attributes-> Drag and Drop attributes from leftmost pane of attributes to middle pane of Hierarchy.

Drag fields in sequence from Attributes to Hierarchy window (Year, Quarter Name, Month Name, Week of the Month, Full Date UK),

Step 8: Deploy the Cube

8.1 In Solution Explorer, right click on Project Name -- > Click **Properties**

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8.2 Set **Deployment Properties** First. In Configuration Properties, Select Deployment-> Assign Your SQL Server Instance Name Where Analysis Services Is Installed (*mubin-pc\fairy*) (*Machine Name\Instance Name*) -> Choose Deployment Mode **Deploy All** as of now ->Select Processing Option **Do Not Process** -> Click **OK**

8.3 In Solution Explorer, right click on **Project Name** (SalesDataAnalysis) -- > Click **Deploy**

8.4 Once Deployment will finish, you can see the message **Deployment Completed** in deployment Properties.

Step 9: Process the Cube

9.1 In Solution Explorer, right click on Project Name (SalesDataAnalysis) -- > Click **Process**

9.2 Click on **Run** button to process the Cube

9.3 Once processing is complete, you can see **Status** as **Process Succeeded** -->Click **Close** to close both the open windows for processing one after the other.

Step 10: Browse the Cube for Analysis

10.1 In Solution Explorer, right click on Cube Name (SalesDataAnalysisCube) -- > Click **Browse**

10.2 Drag and drop measures in to Detail fields, & Drag and Drop Dimension Attributes in Row Field or Column fields.

Now to **Browse Our Cube**

1. Product Name Drag & Drop into Column
2. Full Date UK Drag & Drop into Row Field
3. FactProductSalesCount Drop this measure in Detail area

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